



REDUCED vs. STANDARD DOSE APIXABAN FOR SECONDARY PREVENTION OF CANCER-ASSOCIATED VENOUS THROMBOEMBOLISM: A POOLED ANALYSIS

Vasu Malhotra, D.O.¹, Shreya G. Patel, D.O.¹, Ardit Feinaj, M.D.¹, Devesh Amin, M.D.¹, Henry Ash, D.O.¹, Mohammad B. Booze, M.D.², Zachary Breslow, D.O.¹, Jennifer Trube, D.O.¹, Muhammad Zain Farooq, M.D.³, Michael Sabina, D.O.¹

¹Lakeland Regional Health, Department of Internal Medicine, Lakeland, FL, USA; ²Southern Illinois University School of Medicine, Department of Internal Medicine, Springfield, IL, USA; ³Lakeland Regional Health, Department of Hematology & Oncology, Lakeland, FL, USA



INTRODUCTION



Cancer-associated venous thromboembolism (VTE) requires prolonged anticoagulation due to persistent thrombotic and bleeding risks from malignancy and chemotherapy.



While full-dose apixaban (5 mg twice daily) is recommended for secondary prevention, extended use may increase bleeding risk in cancer patients.



Reducing the dose to 2.5 mg twice daily after the initial six months has been evaluated as a strategy to lower bleeding risk while maintaining efficacy. This is the first pooled analysis that combines the two available randomized trials to address the uncertainty surrounding optimal long-term apixaban dosing for cancer-associated VTE.

AIM



To compare the efficacy and safety of reduced-dose (2.5 mg BID) versus standard-dose (5 mg BID) apixaban for secondary prevention of cancer-associated VTE after at least six months of full-dose anticoagulation.

METHOD

Study Design

- Meta-analysis following PRISMA guidelines
- PROSPERO: CRD42025102633
- Random-effects model

Search Strategy

- 5 databases searched - November 2025
- PubMed, Embase, CENTRAL, ClinicalTrials.gov, Web of Science
- 1,358 records screened

Eligibility

- RCTs comparing reduced-dose (2.5 mg BID) vs standard-dose (5 mg BID) apixaban
- Cancer patients with VTE after ≥6 months anticoagulation
- Outcomes: recurrent VTE, bleeding events, mortality

Statistical Analysis

- Pooled hazard ratios (HR) and risk ratios (RR) with 95% CI
- Heterogeneity assessed using I² statistic
- Risk of bias: RoB 2.0; Certainty: GRADE framework

RESULTS

Included Studies

Trial	EVE 2024	API-CAT 2025	POOLED
Design	RCT, double-blind, multicenter	Phase III RCT, double-blind, multicenter	
Total N	360	1,766	2,126
Reduced Dose	179	866	1,045
Standard Dose	181	900	1,081
Follow-up	12 months	12 months	12 months

0-6 Months
Standard-dose phase
(5 mg BID)

6-18 Months
Reduced-dose phase
(2.5 mg BID)



Population: Patients with active cancer who completed ≥6 months anticoagulation for VTE



Median time since index VTE: 8 months (IQR 6-12)

Primary Findings

- The composite of recurrent VTE with bleeding showed a lower risk (RR 0.79, 95% CI 0.65–0.96; I² = 0%). The composite of major and clinically relevant nonmajor bleeding was also reduced (HR 0.75, 95% CI 0.63–0.88; I² = 0%).
- No significant differences were observed for recurrent VTE (HR 0.83, 95% CI 0.16–4.29; I² = 0%), major bleeding (HR 0.72, 95% CI 0.05–11.13; I² = 0%), clinically relevant nonmajor bleeding (HR 0.77, 95% CI 0.36–1.63; I² = 0%), all-cause mortality (HR 0.96, 95% CI 0.66–1.40; I² = 0%), deep-vein thrombosis (RR 0.95, 95% CI 0.44–2.08; I² = 0%), or pulmonary embolism (RR 0.89, 95% CI 0.43–1.87; I² = 0%).

Quality of Evidence

Heterogeneity

- I² = 0% across all outcomes (complete consistency between trials)

Risk of Bias

- Low to moderate (Cochrane RoB 2.0)

Certainty of Evidence

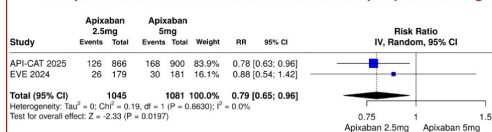
- High to moderate (GRADE framework)

Statistical Model

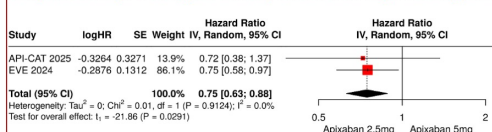
- Random-effects with Knapp-Hartung adjustment

Primary Findings

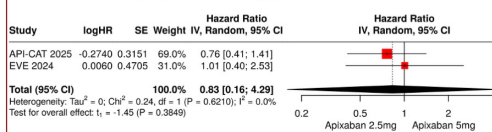
Composite of Recurrent Venous Thromboembolism (VTE) and Bleeding



Composite of Major Bleeding and Clinically Relevant Non-major Bleeding (CRNMB)

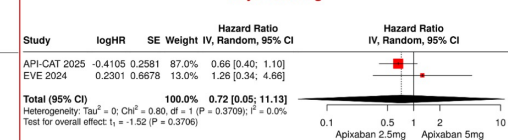


Recurrent Venous Thromboembolism (VTE)

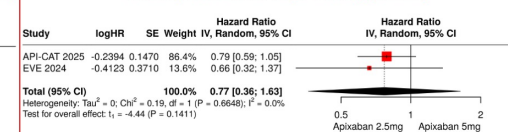


Note: RR or HR < 1.0 favors reduced-dose apixaban (2.5 mg BID)

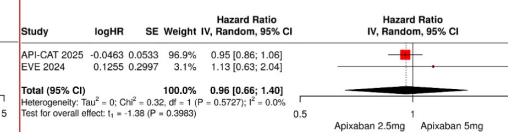
Major Bleeding



Clinically Relevant Non-major Bleeding (CRNMB)



All Cause Mortality



CONCLUSIONS



In this pooled analysis of 2,126 patients indicate that reduced-dose apixaban provides comparable protection against recurrent thromboembolism relative to the standard dose. Rates of recurrent VTE, deep-vein thrombosis (DVT), pulmonary embolism (PE), and all-cause mortality were statistically similar between treatment arms, suggesting no meaningful difference in efficacy.



Both composite outcomes favored reduced-dose apixaban. The composite of recurrent VTE with bleeding occurred 21% less often (RR 0.79, 95% CI 0.65–0.96), and the composite of major or clinically relevant non-major bleeding was 25% lower with the 2.5 mg dose (HR 0.75, 95% CI 0.63–0.88).



Interpretation is limited by: (1) the dominance of the larger API-CAT trial and the small number of available RCTs (2) both trials enrolled patients who had completed at least six months of anticoagulation without complications, which limits applicability (3) long-term outcomes beyond 12 months remain unassessed



Large, multicenter, prospective studies with extended follow-up are needed to assess whether reduced-dose apixaban remains safe and effective beyond the 12-month horizon. Future studies should evaluate outcomes in populations underrepresented in our data, including those receiving intensive chemotherapy, patients with hematologic cancers, and those with significant renal impairment.

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CONTACT & REFERENCES

Name: Vasu Malhotra, D.O.
Address: 1324 Lakeland Hills Blvd, Lakeland, FL 33805, USA
Email: Vasu.Malhotra@myLRH.org
Twitter: @oncosaurus

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